STU001 · Q1

Inserting into a balanced BST takes O(log n) time. The tree height is log(n) because the balancing invariant (such as AVL or Red-Black) keeps the height bounded. To insert, we compare the new key with the current node, recurse into the left subtree if smaller or right subtree if larger, until we reach a leaf where we attach the new node. After insertion, rebalancing via rotations takes O(1) work per rotation and is amortized O(1) overall.